

RED RIVER COLLEGE POLYTECHNIC · WORKSHOP

Digital World Building

Consider → Scope → Build: from an idea to an explorable world

Dann Blair · Founder & CTO, BSD XR



// HELLO, I'M DANN

Daniel Blair

Educator · Researcher · Technologist

FOUNDER & CTO	BSD XR — applied AI + spatial computing
INSTRUCTOR	ACE Project Space, RRC Polytech
BOARD CHAIR	New Media Manitoba

Construction | Aerospace | Mining | Defence | Utilities

Award-winning entrepreneur and developer. Technology with purpose.

My path: developer, founder, educator

Build

Hands-on developer first, with roots that predate the studio

Found

Started BSD XR; Clutch Top 15 Global AR/VR Developer and Top Unity Developer 2024

Share

TEDx Winnipeg · MIT Reality Hack panelist · Future Leaders of Manitoba Award

Teach

Instructor, ACE Project Space, RRC Polytech · Board Chair, New Media Manitoba

I don't vibe code. I call it **AI co-authoring**: AI drafts, I stay the author, guided by the 4 Ds.

TODAY

Foundations first, then we build

Part 1 · Foundations

What are digital worlds?
The metaverse
Digital storytelling
Digital identity & ownership
BSD XR: the worlds we build

Part 2 · Workshop

Consider — the decisions behind a world
Scope — write it down, by hand and with AI
Tools — engines, Blender + Claude, ComfyUI
Build — best practices, then create your world
Playtest & wrap

01 · WHAT ARE DIGITAL WORLDS?

A digital world is a space you can be in — with its own rules, its own story, and a reason to be there.

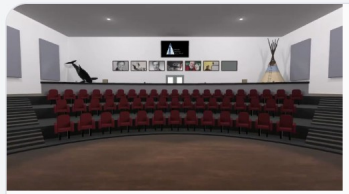
Not a website you read or a video you watch. A place you move through, act in, and come back to.

Worlds come in many kinds



Games

Play, challenge, adventure



Social spaces

Meet, gather, perform



Training simulations

Practise the dangerous safely



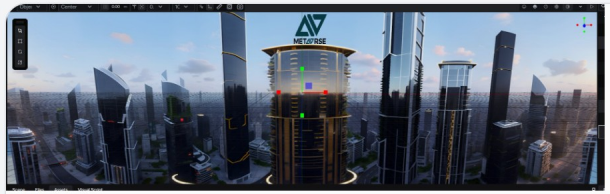
Digital twins

Mirror a real place or system



Museums & heritage

Preserve and share the past



The spatial web

Worlds behind a link

Five ingredients every world shares



Space

Somewhere to be, with scale and layout



Presence

The feeling that you are there



Rules

What's possible, and what happens when you act



Story

A history and a reason to explore



Purpose

Why it exists and who it's for

Miss one and people feel it, even if they can't name it.

02 · THE METAVERSE

An idea from science fiction, 30 years in the making

1992

Snow Crash by Neal Stephenson coins "the Metaverse"

2003

Second Life: avatars, land and a user-built economy

2006

Roblox launches: players become world builders

2017

Fortnite: a game that becomes a venue for concerts and events

2021

Facebook renames itself Meta. Peak hype

2024

Apple Vision Pro brands it "spatial computing"

What people mean by ”the metaverse”

Persistent

It keeps existing when you log off

Real-time 3D

Rendered live, explored freely

Shared

Many people, at the same time, in the same place

Present

Each person feels like they're there

Interoperable

Worlds connect; you can move between them

Continuous

Your identity, items and history come with you

Adapted from Matthew Ball, *The Metaverse: And How It Will Revolutionize Everything* (2022)

XR · EXTENDED REALITY

XR is every way of stepping into a world

The real world — A fully virtual world

AR

AUGMENTED REALITY

Digital layered onto the real world: phone, tablet, glasses

MR

MIXED REALITY

Real and virtual share the room and react to each other

VR

VIRTUAL REALITY

You step fully inside; the real world disappears

After Milgram & Kishino's reality–virtuality continuum (1994)

XR takes the metaverse off the screen



AR · THE WORLD COMES TO YOU

Fossil Discovery Adventure

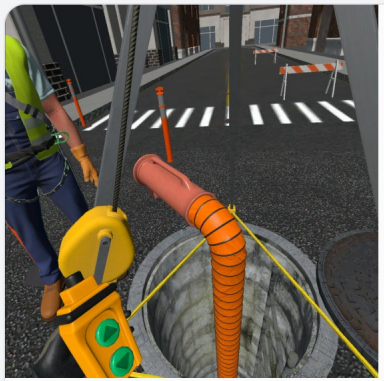
A life-size mosasaur appears inside the museum in Morden



MR · YOUR BODY IS THE CONTROLLER

The Grid: Near Miss

Passthrough training: throw a real switch, look both ways



VR · YOU GO TO THE WORLD

Confined Space VR

A silo, a sewer, a crawl space, with no real risk

What XR changes for world builders

Scale is felt

A door that's 10 cm too short feels wrong instantly

The body is the input

Hands, head, gaze and voice replace buttons

The room is part of the level

Real walls, tables and people shape what fits

Comfort is a requirement

Teleport, steady frame rate, no forced camera moves

There is no one metaverse — there are many worlds, and the ones that last have a job

What was hyped

Buying virtual land as an investment

One platform for everything

Everyone working in VR all day

What stuck

Training and simulation

Games as social places

Digital twins, museums and education

3D on the open web

03 · DIGITAL STORYTELLING

In a world, the space is the story.

Film tells you what happened. A world lets you find out.

Four ways a space tells a story

Evoked

Borrows a story you already know: a haunted house, a frontier town

Enacted

You perform the story: the quest, the rescue, the climb

Embedded

Clues placed in the space: a journal, a broken door, an empty chair

Emergent

Stories that arise from rules and players, not a script

Henry Jenkins, "Game Design as Narrative Architecture" (2004)



STORY THROUGH MECHANICS

Every word brings you closer to home

Reclaim! Azhe-giiwewining — a point-and-click adventure where learning Ojibwe words is how Miskwaa finds her way back to her grandmother.

With Grassroots Indigenous Multimedia · on Steam

STORY THROUGH CONSEQUENCE

Forget to look both ways, and the train arrives

The Grid: Near Miss for CN Rail — mixed-reality training where you throw a real switch, "Point and Call", and live with the result.



04 · DIGITAL IDENTITY & OWNERSHIP

Who are you in a world?

Avatar

How others see you — chosen, scanned or assigned

Presence

How it feels to be you, there — voice, hands, gaze

Persistence

What the world remembers about you between visits

The Proteus effect: people tend to behave like their avatar looks. Design choices shape behaviour.

Yee & Bailenson, "The Proteus Effect" (2007)

What do you actually own in a world?

What it feels like you own	What you usually have
A skin or item you paid for	A licence, under the platform's terms
A world or object you built	Rights that depend on the platform's terms
Your avatar, voice or 3D scan	A likeness you may have licensed to the platform
Your behaviour in the world	Data the platform collects — movement, gaze, voice

NFTs promised portable digital ownership; the 2021 boom faded fast in 2022. The question they raised — **can you leave with what's yours?** — is still open.

OWNERSHIP OF KNOWLEDGE

When a world holds a community's knowledge, who controls it?

- **O**wnership
- **C**ontrol
- **A**ccess
- **P**ossession

OCAP® — First Nations principles for how data and knowledge are governed



OCAP® is a registered trademark of the First Nations Information Governance Centre (FNIGC) · Image: Myera Farms digital twin

Five questions every world builder should answer

1. Who owns what people make here?
2. What data do we collect, and why do we need it?
3. Can people leave with what's theirs?
4. Whose culture, place or likeness are we using — and who said yes?
5. How do people choose how they appear — and stay safe?

// TRADES & SAFETY

Built in Manitoba, used on the job



CN Rail · Near Miss

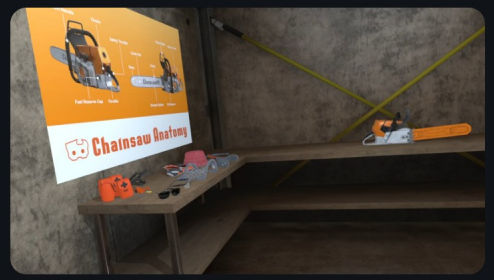
Mixed reality: throw a real switch, Point and Call, gaze-validated safety checks



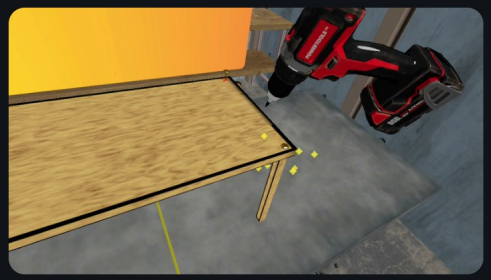
CN Rail · Working at Heights



MadeSafe · Confined Space



MCSC + WCB · Chainsaw VR



MCSC · Power Drill VR

CASE · CN RAIL

One world, many lessons

The Grid is a family of training modules — Working at Heights, Switches & Routes, Near Miss, Slack Simulator — built on shared rail environments.

World-building lesson: build a modular kit once, then reuse it for every scenario.







CASE · CEDF

Bringing northern museums to anyone

A VR museum of northern Manitoba history, with 3D-scanned artifacts placed in custom exhibit worlds.

Lesson: capture the real thing, then design the space around it.

Worlds for gathering and belonging



VWS Amphitheater

For the Virtual World Society on ENGAGE: a stage, breakout tables and a wall that tells the society's history.



Hawaiian Cultural Worlds

Native flora and fauna, labelled in Hawaiian and English — a living encyclopedia to explore.

// WHAT WE'VE LEARNED

What good immersive worlds get right

01	Start from a real problem, incident or near miss, not the technology
02	Build it with the people who live it
03	Measure what people did, not just that they finished
04	Keep it short enough to repeat
05	Run it where people live and work

PART 2 · THE WORLD SEED DEMO

Two ways into a world.

01 · LEARN

Developing for WebXR

How a world gets built for the browser and the headset

02 · CREATE

Build your own world

Design it and view it in the browser. No code.

Plus an optional challenge: take the repo, extend the tool, build something bigger.

Ten questions every world must answer

Consideration	The question
Purpose & audience	Why does it exist, and who is it for?
Platform & constraints	Browser, phone, headset? What does that cost us?
Experience & verbs	What does a visitor do in the first 60 seconds?
Space & wayfinding	How is it laid out, and how do people find their way?
Story & lore	What happened here before you arrived?
Rules & systems	What's true here — time, weather, physics, economy?
Look, sound & mood	Three mood words, five colours, one sound
Comfort & access	Motion comfort, seated play, captions, contrast
Culture & ethics	Whose places, symbols and stories? Who said yes?
Scope & feasibility	What can we build with our time, skills and budget?

SCOPE

A world is scoped in layers

Each layer is only as detailed as the build needs.

1 · World Brief	One page: purpose, pillars, experience
2 · World Bible	Lore, rules, tone, naming
3 · Spatial Plan	Map, zones, paths, landmark, scale
4 · Asset List	Every object, its source and budget
5 · Interaction Spec	Trigger → response
6 · Build Plan	Milestones and what "done" means

Two paths to a brief: by hand, or with AI and the 4 Ds

Step	By hand	With AI
Frame	Whiteboard the ten questions	Direction: set goal, audience, limits first
Generate	Sticky notes, sketches, mood board	Description: a prompt chain, not one prompt
Judge	Group critique against pillars	Discernment: cut the generic and the wrong
Hand off	Brief goes to the team	Delegation: AI drafts; humans own taste

AI is fast at volume and structure. **You** own the pillars, the culture and the final call.

ACTIVITY · 25 MINUTES

Brief it twice

10 min

By hand

Write a one-page World Brief for your world idea

10 min

With AI

Ask AI to interview you, draft the brief, then critique it

5 min

Compare

What did AI add that's good? Generic? What only you knew?

Output: one revised World Brief per group. You'll build it in the browser after the break.

Break · 10 minutes

Next: we build one of your worlds, live.

TOOLS · GAME ENGINES

Four ways to bring a world to life

	Godot	Unity	Unreal	Web (A-Frame)
Cost	Free, open source	Free under $200K	5% after $1M	Free, open source
Language	GDScript, C#	C#	Blueprints, C++	HTML, JavaScript
Strength	Light, gentle start	XR and mobile reach	Top-end visuals	Runs from a link
Watch for	XR still maturing	Terms have changed	Heavy hardware	Performance ceiling

Licence terms as of fall 2026 — check before you commit. Unity: Personal plan under $200K revenue or funding. Unreal: per title, lifetime gross.

Let the brief choose the engine

"It has to open from a link, on any device"	→ **Web, or Godot**
"It has to run on a standalone headset"	→ **Unity**
"Photoreal is one of our pillars"	→ **Unreal**
"Nobody on the team codes yet"	→ **Godot, or Blueprints**

`> Problem → People → Process → Tool. The tool comes last.`

TOOLS · ASSET PIPELINE

From an idea to an asset in your world

01
Asset list
What, where, how many triangles

→

02
ComfyUI
Concept image, texture, or a rough mesh

→

03
Blender + Claude
Build, clean up, scale, name, export

→

04
Engine
GLB in, test at true scale

AI speeds up every step. **The asset list still decides what gets made.**



A winter forest XR world, generated with AI

// WHY I'M OPTIMISTIC

AI makes the first version cheap again

THEN	"Show me it's possible."
NOW	"Show me it's finished."

Don't skip the step where you learn. Use cheap AI drafts to greybox faster, not to skip the greybox.

LIVE DEMO · BLENDER + CLAUDE

Describe it in words, watch it appear in Blender

How it works

The **Model Context Protocol (MCP)** connects Claude to Blender. Claude reads your scene, writes Blender Python, runs it, and checks the result.

What we'll do

1. Build a lantern from the asset list
2. Fix scale, pivot and names
3. Export a GLB
4. Import it into the World Seed editor

Official Blender Lab MCP server (Blender 5.1+) · community "MCP for Blender" adds Poly Haven, Sketchfab and AI model generation

LIVE DEMO · COMFYUI

One node graph: images, video and 3D

Images

Concept art, mood boards, textures and skyboxes

Video

Animatics, mood clips, trailer beats

3D

Image to textured mesh, then clean-up in Blender

Runs on your own GPU. Every step is a visible node, so you can see and share the whole recipe.

Four guardrails for AI-made assets

Check the model's licence

Not just the output: some models restrict commercial use

Ask before you generate

No real people, logos or cultural designs without permission

Label it

Mark AI-generated assets in your licence ledger

Clean it up

Fix scale, triangle count, holes and UVs before it ships

BUILD

Build in stages — greybox first, polish last

1	2	3	4	5
Greybox	**Blockout**	**Art pass**	**Life**	**Polish**
Prove layout, scale and flow with boxes	Zones, paths, landmarks, sightlines	Palette, style, lighting from the brief	Interaction, audio, story beats	Fog, particles, tuned to budget

Changing a box costs seconds. Changing a finished building costs days.

Habits that keep worlds buildable

Real scale

1 unit = 1 metre. Doors 2.1 m, eyes 1.6 m

Wayfinding

A landmark visible from everywhere; light leads the eye

Modular kits

Reusable pieces, shared pivots, snap to grid

Budgets

Set triangle, draw-call and frame-rate targets up front

One style

Consistency beats fidelity

Licence ledger

Log the source and licence of every asset

Commit small

Version control from day one

Scope = truth

When world and doc disagree, fix one on purpose

YOUR TURN · WORLD SEED: CREATE

Build your world in the browser. No code.

1

Greybox

Block out zones, paths and the landmark with plain shapes

2

Dress it

Add objects, colours and your own GLB models

3

Light it

Set the mood your brief describes

4

Save & view

Save it, then view your world in the browser

Your World Brief is the plan. Every object should answer to it.

// HOW WE'LL USE AI TO BUILD

Vibe coding vs AI co-authoring

	Vibe coding	AI co-authoring
Who's the author?	The vibes	You are. Your name is on it.
The output	Forget it exists	Read it, test it, own it
The goal	Make it look done	Make it right for here
Best for	Throwaway experiments	Anything someone relies on

Rule of thumb: if your name is on it, you read all of it.

Experienced developers using AI were 19% slower on real tasks, yet believed they were 20% faster. METR randomized trial, July 2025

OPTIONAL CHALLENGE

Extend the tool. Grow the world.

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Take the repo

Fork World Seed from GitHub and run it on your machine

Extend the tool

Add a feature your world needs: a new object, an interaction, sound

Build bigger

A larger, multi-zone world, published to a link

Use Claude Code as a co-author. **Your name is on it, so read every line.**

github.com/CMDann/digital-world-building

PLAYTEST · 15 MINUTES

Walk it, then give three notes

One that reads

What did you understand without being told?

One that confuses

Where did you get lost or stuck?

One to cut

What would you remove to make it stronger?

The scope is the product before the product

Consider

Ten questions, answered on purpose

Scope

Written down, by hand or with AI

Build

In stages, to budget, playtested

Take home: the ten questions, the 4 Ds, and a World Brief of your own.



Go build a world.

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